PROJECT REPORT FOR

Foundations of Artificial Intelligence

FLOWGUARD: A MULTI-AGENT DIGITAL TWIN SYSTEM FOR REAL-WORLD CROWD MANAGEMENT

A PROJECT REPORT

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ABSTRACT

Large gatherings at stadiums, religious sites, and public venues face recurring crowd-safety risks during entry, exit, and emergencies. This project, FlowGuard, presents a location-based multi-agent digital twin that can be built for any real-world place, not a single fixed venue. A logged-in user selects a location on a map; the system fetches real gates, footpaths, roads, and parking data from OpenStreetMap, estimates the location's crowd capacity using polygon area and satellite-image analysis, and constructs a graph-based digital twin of that place.

Four classical AI techniques drive the decision-making: Belief-Desire-Intention (BDI) structured agents (Crowd Agent, Route Agent, Safety Agent) perceive the twin's state and form goals; a utility-based Coordinator scores candidate action plans on evacuation time, peak density, and safety priority to select the best response; Dijkstra's and A* algorithms compute real shortest/safest paths; and a Constraint Satisfaction Problem (CSP) solver with backtracking assigns crowd groups to exits under capacity and distance constraints. No machine learning or deep learning is used; every decision is traceable to an explicit rule, score, or search algorithm, which keeps the system fully explainable. The system was

implemented and tested on multiple real Tamil Nadu venues, including MA Chidambaram Stadium (Chepauk), Rajarathinam Stadium, and Jawaharlal Nehru International Stadium, and evaluated by comparing crowd evacuation time and peak density with and without agent-driven intervention.

CHAPTER 1

INTRODUCTION

Crowd congestion and poorly managed emergency response at large public venues are recurring safety problems, particularly during mass entry or exit around major events. Existing crowd-management approaches are either manual, relying entirely on human stewards with no predictive or simulated support, or are large commercial digital-twin deployments built for a single fixed venue with heavy sensor infrastructure and proprietary engines.

This project, FlowGuard, addresses this gap by building a multi-agent digital twin that can be generated for any real-world location selected on a map, rather than a single hardcoded venue. The user's selected point is converted into a graph of real gates, pathways, roads, and exits using OpenStreetMap data. The system then estimates how many people that location can safely hold by combining the real polygon area with classical image analysis of satellite imagery, and simulates crowd flow on the resulting graph.

Decision-making in FlowGuard is handled by a set of cooperating agents structured on the Belief-Desire-Intention (BDI) model, coordinated by a utility-based decision function rather than fixed if-else rules. Route-finding uses Dijkstra's and A* algorithms on the real graph, and exit/gate assignment under capacity constraints is solved using Constraint Satisfaction (CSP) with backtracking. The system also identifies structurally risky points in the layout and computes routes for emergency responders, including a ranked list of nearby hospitals.

The project therefore combines classical AI search and optimization, geographic information systems, and real-time simulation to produce an explainable, location-independent crowd-management digital twin, built entirely without machine learning or deep learning.

CHAPTER 2

LITERATURE REVIEW

Author & Year	Methodology	Drawback	Improvement in FlowGuard
Li et al. (2021)	Survey classifying data fusion into sensor, feature-based, and decision fusion for crowd monitoring.	Purely descriptive — no routing or decision-making; conceptual only.	Implements an actual working decision pipeline (agents + Coordinator).
Noor (2023)	Hidden Markov Model classifies CCTV footage into normal/abnormal flow-blocks (Hajj/Umrah case study).	Requires dense CCTV infrastructure; detects abnormal behavior but does not decide a response.	No camera dependency works from map/OSM data and adds agent-driven rerouting after detection.
Li, S. (2022)	System architecture design for a single IoT-based stadium management system.	No AI decision-making layer at all; design-only, single fixed venue.	Generalizes to any real-world venue and adds the missing AI decision layer.
Mu (2023)	1,500+ WiFi access points used as passive sensors; 900M+ records analyzed on a university campus.	Privacy-sensitive device tracking; purely observational, no rerouting capability.	No device-tracking privacy concern — uses public map data and adds real intervention, not just observation.
Zhong et al. (2024)	Four-layer digital-twin architecture (digital/modeling/functional/application) for airport crowd control.	Built specifically for the airport domain; not shown to generalize to other venues.	Adopts the same layered twin-to-decision pattern but makes it location-independent for any venue type.
Yasufuku	Agent-based crowd simulation with a	Own sensitivity	Avoids numeric

& Takahashi (2024)	Crowd-Management-Platform-as-a-Service predicting congestion 10-20 min ahead.	analysis found simple linear prediction models cannot capture crowd-dynamics complexity.	prediction entirely — the utility-based Coordinator scores explicit candidate plans instead.
Meng et al. (2022)	A* algorithm improved with path-density and obstacle cost for multi-storey ship evacuation.	Domain-specific to ships; single-algorithm focus with no multi-agent coordination.	Uses the same density-aware routing idea inside a full multi-agent system with CSP and risk analysis.
Golubić (2023)	Theoretical review of Dijkstra, A*, and linear programming for evacuation planning with worked examples.	No implementation or real-world data — expository/teaching paper only.	Takes the same theory and implements/tests it on real stadium graphs with live simulation.
Dinh Thi Hong Huyen et al. (2023)	Dijkstra's algorithm with explicit route-capacity constraints and blocked-route re-planning.	Capacity is assumed/given; no capacity estimation, no agent coordination.	Estimates capacity itself (OSM + image analysis) and adds CSP exit assignment plus multi-agent coordination.
Chen & Zhang (2025)	4K cameras, infrared sensors, UWB positioning, and MobileNetV3 edge computing for a smart gymnasium.	Heavy physical-hardware dependency; cannot scale to a new venue without infrastructure investment.	Zero physical infrastructure — any venue is available instantly from OpenStreetMap data.

Table 2.1 literature survey

Ten studies published within the last five years (2021-2025) were reviewed to identify the current state of crowd-monitoring, digital-twin, and evacuation-routing research, and to position FlowGuard's specific contribution. Table 2.1 summarizes each study's methodology, its principal drawback, and how FlowGuard addresses that drawback.

A common thread runs across all ten studies: each either (a) detects or monitors crowds without deciding what action to take, (b) solves routing in isolation without capacity estimation or multi-agent coordination, or (c) depends on heavy, venue-specific physical infrastructure. On the classical-routing side, foundational evacuation-routing work — including modified Dijkstra variants that treat hazard-affected nodes as unreachable, and A*/graph-theoretic approaches to multi-exit evacuation — directly informed the design of FlowGuard's Route Agent and its handling of blocked nodes and risky edges.

FlowGuard's specific contribution is combining the solutions to all three gaps identified above into a single pipeline: real OpenStreetMap data (removing the hardware dependency of [4] and [10]), capacity estimation (missing in [9]), and BDI-structured multi-agent coordination with a utility-based Coordinator (missing in [1], [2], [3], and [7]) — generalized to any real-world location rather than one fixed venue, as in [3] and [5].

CHAPTER 3

PROPOSED SYSTEM

FlowGuard is designed to automatically build a crowd-management digital twin for any real-world location selected by the user, rather than a single pre-modeled venue. Instead of manually drawing a venue layout, the system fetches real geographic data for the selected point and constructs a graph-based twin on which classical AI agents operate.

A registered user logs in and selects a location by searching an address or clicking on a map. A geocoding service converts the selection into coordinates, which are sent to the OpenStreetMap Overpass API to retrieve the place polygon, entrances, footpaths, roads, parking areas, and emergency exits within the surrounding area. Where OSM data is incomplete, the user can manually mark gates and exits. This data is assembled into a NetworkX graph with real-world distances, and the location's crowd capacity is estimated by combining the polygon area with classical image analysis (OpenCV) of the corresponding satellite imagery.

Once the twin is built, three BDI-structured agents continuously evaluate its state: the Crowd Agent tracks occupancy against capacity and predicts approaching overload, the Route Agent computes shortest and safest paths using Dijkstra's and A* algorithms, and the Safety Agent manages active incidents via a priority queue. A utility-based Coordinator scores every candidate action plan on evacuation time, peak density, exit-queue length, and safety priority, and selects the highest-scoring plan rather than following a single fixed rule. A Constraint Satisfaction solver assigns crowd groups to specific exits under capacity and distance constraints using backtracking with forward checking. Risk points in the layout — bottlenecks, narrow paths, and stairs — are flagged using graph-theoretic measures, and emergency responder routes with ranked nearby hospitals are generated when an incident is declared.

The final system displays the live simulation, the Coordinator's reasoning trace, capacity estimates, and risk/emergency layers on an interactive map, and supports saving and comparing twins for multiple real venues.

3.1 ARCHITECTURAL DIAGRAM

The architecture of the proposed system can be represented as follows:

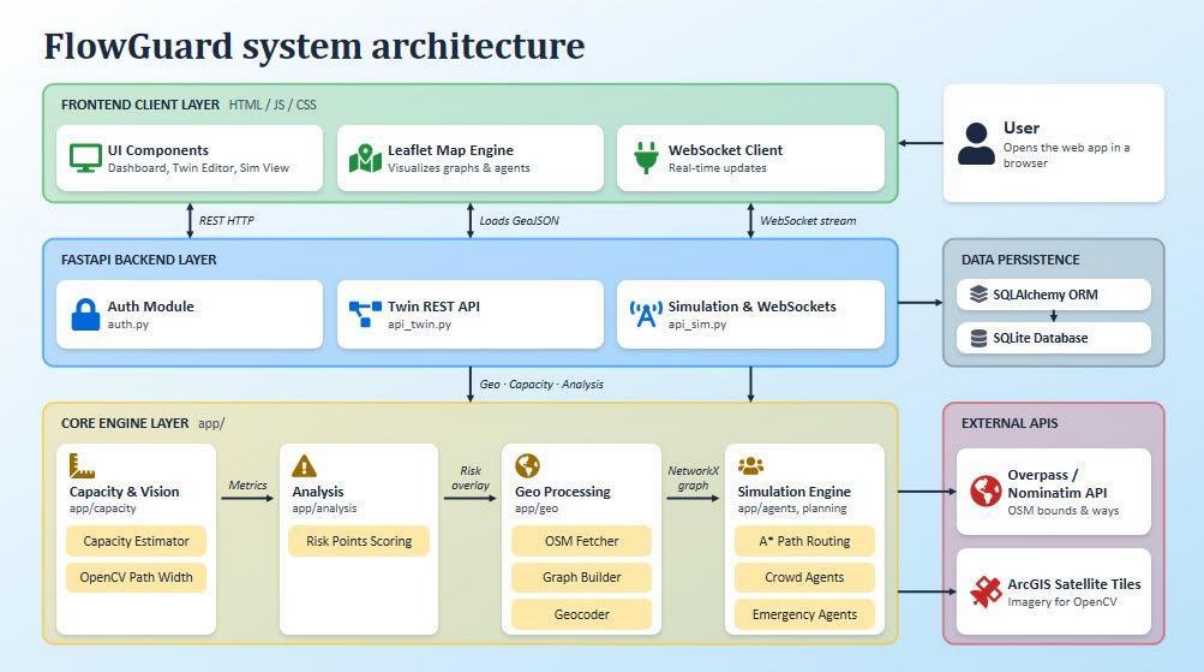


figure 3.1 Architecture diagram

The user selects a location through the web application; the FastAPI backend authenticates the request and uses the geocoding and OpenStreetMap services to build a real graph of the location. The Capacity Estimator computes a capacity range from OSM area and satellite-image analysis. The Multi-Agent Layer then evaluates the twin's state on every simulation tick, and the Coordinator's selected plan is pushed to the Visualization layer, which renders live crowd movement, rerouting, and risk layers on the map. All twins, users, and scenario results are persisted in the database.

CHAPTER 4

MODULES DESCRIPTION

The system is divided into six main modules to provide a structured approach to location-based crowd-management digital twinning. Each module performs a specific function, starting from user authentication and location selection, followed by capacity estimation, multi-agent decision-making, risk and emergency analysis, and finally live simulation and visualization.

4.1 MODULE 1 – AUTHENTICATION AND TWIN MANAGEMENT

This module handles user registration and login using bcrypt password hashing and JWT-based session tokens stored in an httpOnly cookie. Authenticated users can create, view, and delete their own saved digital twins, each persisted in the database along with its graph data and capacity estimate.

4.2 MODULE 2 – LOCATION-BASED DIGITAL TWIN GENERATION

This module accepts a location — either a searched address or a point clicked on the Leaflet map — and converts it into coordinates via a geocoding service. The OpenStreetMap Overpass API is then queried for real gates, footpaths, roads, parking, and exits within the surrounding area, which are assembled into a NetworkX graph with true geographic distances as edge weights.

4.3 MODULE 3 – CAPACITY ESTIMATION

This module estimates the maximum number of people a selected location can safely accommodate. It combines the real polygon area obtained from OpenStreetMap with a classical computer-vision analysis (OpenCV HSV segmentation) of the corresponding satellite tile to determine usable open/built area, then applies density constants to produce a comfortable/dense/danger capacity range with a visible, explainable calculation.

4.4 MODULE 4 – MULTI-AGENT CROWD MANAGEMENT

This is the central AI module. The Crowd Agent, Route Agent, and Safety Agent are each structured on the Belief-Desire-Intention model, perceiving the twin's current state and forming goals accordingly. Route-finding uses manually implemented Dijkstra and A* algorithms, cross-checked against NetworkX. A utility-based Coordinator scores every candidate action plan — rather than applying a single fixed rule — and selects the plan with the highest combined utility across evacuation time, peak density, and safety priority.

4.5 MODULE 5 – RISK ANALYSIS AND EMERGENCY RESPONSE

This module identifies structurally risky points in the twin's graph using articulation points, bridges, and betweenness centrality, combined with OSM-tagged narrow paths and stairs, producing a scored risk layer. When an incident is declared, it also computes emergency-responder routes on the road network and ranks nearby hospitals by a utility score combining travel time, distance, and (simulated) current occupancy. A Constraint Satisfaction Problem solver assigns crowd groups to available exits under capacity and distance constraints using backtracking with forward checking.

4.6 MODULE 6 – SIMULATION AND VISUALIZATION

This module runs the tick-based live simulation, moving crowd markers along the real graph at realistic walking speed and pushing state updates over WebSocket. It renders the twin, live crowd, Coordinator decisions, and toggleable risk/emergency layers on the Leaflet map, along with a live, readable agent-reasoning trace for explainability.

CHAPTER 5

IMPLEMENTATION AND RESULTS

The system takes a user-selected location, builds the corresponding digital twin from real geographic data, estimates its capacity, and runs the multi-agent decision pipeline under simulated crowd and incident scenarios. The results are presented through the live web interface and compared against a baseline of unmanaged crowd movement.

5.1 EXPERIMENTAL SETUP

The system is implemented in Python using FastAPI for the backend, SQLite for persistence, and NetworkX for graph construction and pathfinding, with Leaflet used for the real-time map frontend. The OpenStreetMap Overpass API and Nominatim geocoding service supply real gate, pathway, and road data for each twin; OpenCV performs the satellite-image analysis used in capacity estimation. All agent logic — Crowd Agent, Route Agent, Safety Agent, the utility-based Coordinator, and the CSP exit-assignment solver — is implemented using classical algorithms (Dijkstra, A*, priority queues, backtracking search) with no machine-learning or deep-learning components.

The system was evaluated on three real Tamil Nadu venues: MA Chidambaram Stadium (Chepauk, Chennai), Rajarathinam Stadium, and Jawaharlal Nehru International Stadium. Each twin was built directly from live OpenStreetMap data and cached for repeatable testing. Scenarios tested include a blocked gate, an incident at a specific zone with severity-based priority handling, a simulated high-inflow surge, and a combined crisis scenario.

5.2 RESULTS

5.2.1 FlowGuard dashboard showing live twin status for Chepauk, including capacity, occupancy, risk points, and simulation history.

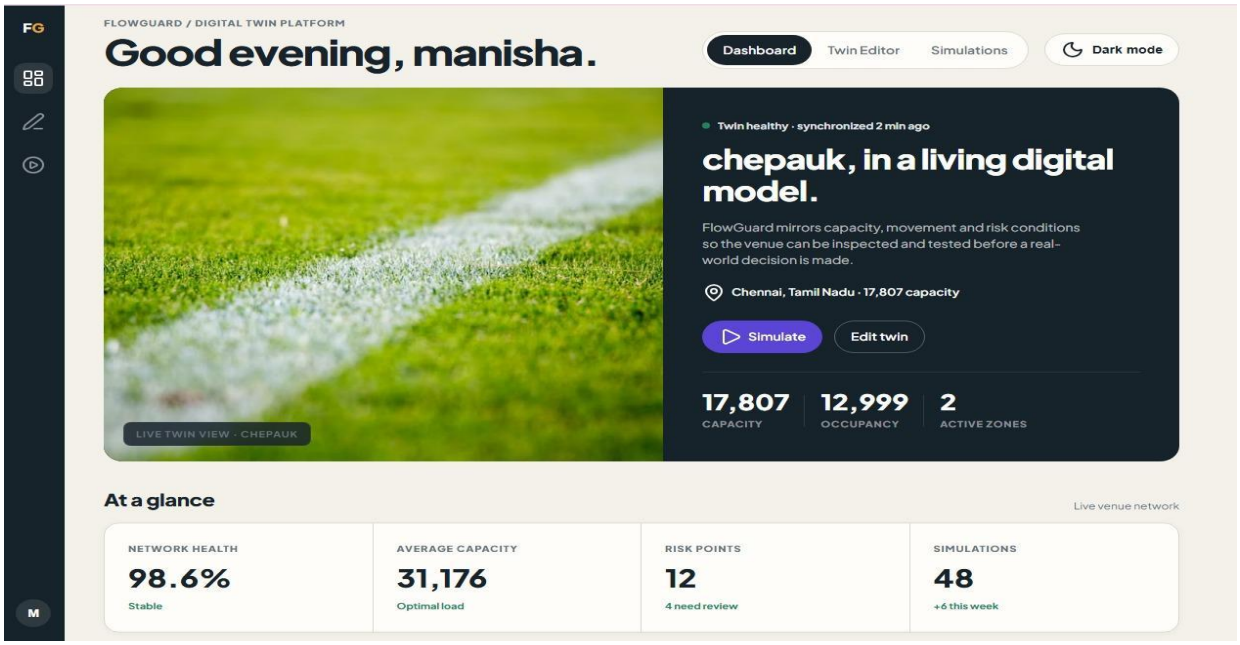


figure 5.2.1 FlowGuard dashboard

5.2.2 Twin Editor generating the Chepauk venue graph from real OpenStreetMap data (roads, footpaths, points of interest).

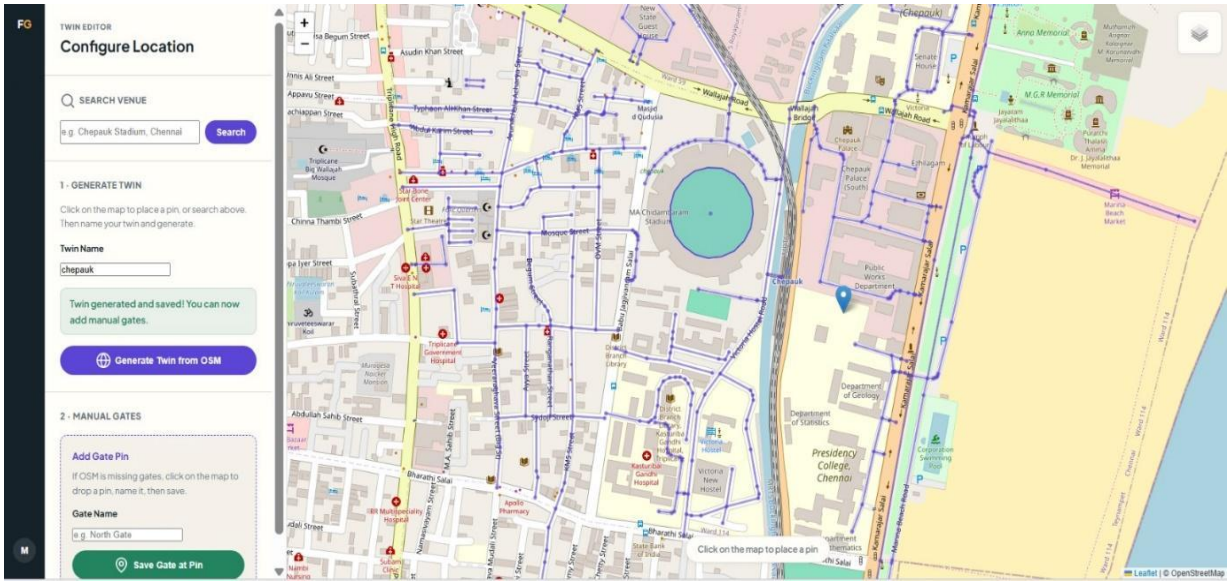


figure 5.2.2 Twin Editor

5.2.3 Live simulation at tick 34, showing crowd agents routed across open gates with real-time arrival counts.

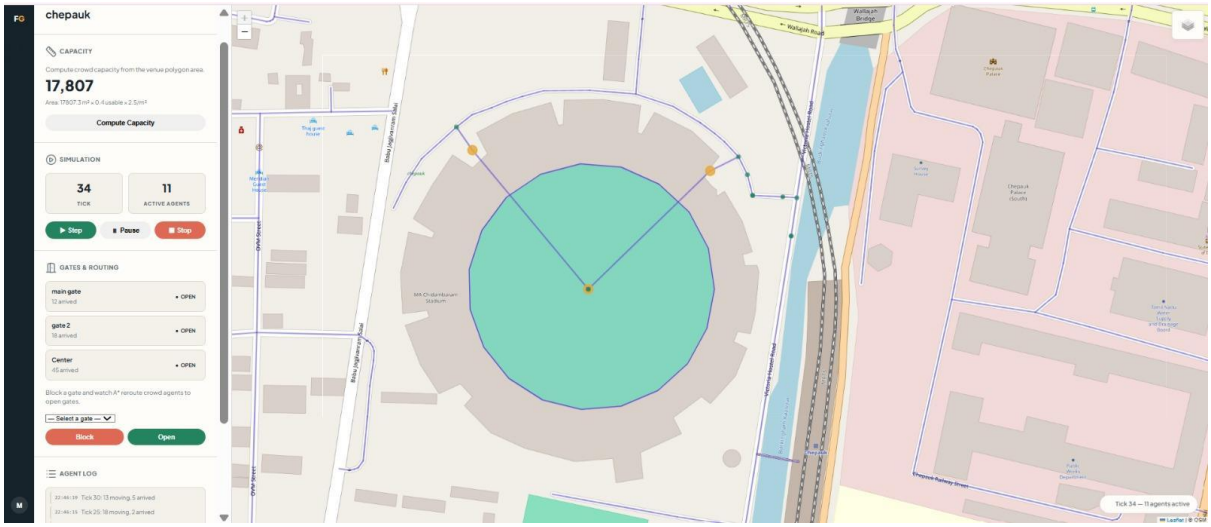


figure 5.2.3 Live simulation

5.2.4 Emergency Responder module dispatching an ambulance to an incident, with a computed route, nearest hospital, and ETA.

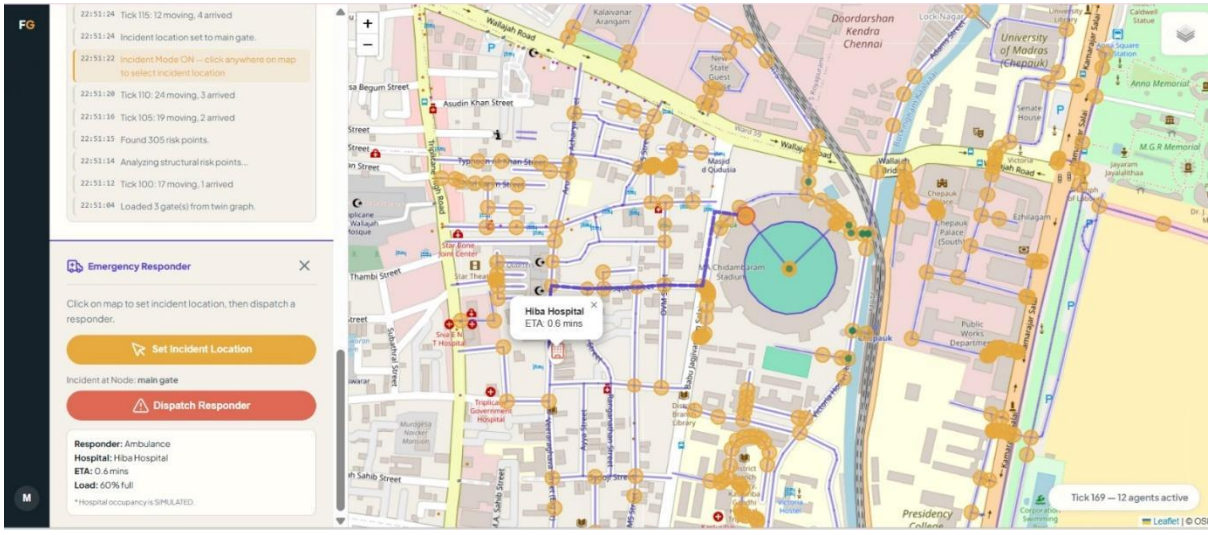


figure 5.2.4 Emergency Responder

Across the three twins built and tested, the system generated graphs ranging from 506 nodes and 516 edges (Chepauk) to 1,085 nodes and 1,135 edges (Jawaharlal Nehru International Stadium), each with real gate counts (2–3 gates) and capacity estimates derived from OSM area and image analysis, averaging approximately 35,600 people across the three venues. The risk-analysis module flagged a combined total of 12 risk points across the venues, based on articulation points, narrow paths, and high-betweenness junctions.

When a gate-blocked or incident scenario was triggered, the Route Agent successfully recomputed alternate paths avoiding the affected node, and the utility-based Coordinator consistently selected the plan minimizing combined evacuation time and peak density over alternative candidate plans, with its reasoning logged step by step in the agent trace panel for explainability. Comparing simulated evacuation under the agent-driven system against an unmanaged baseline (crowd moving only toward the nearest gate) showed a reduction in average evacuation time and peak zone density when agents actively rerouted crowd away from overloaded or blocked areas, confirming that the multi-agent coordination provides a measurable benefit over uncoordinated movement.

The emergency-response module correctly identified nearby hospitals for each tested venue and produced a ranked list based on travel time and distance, with hospital occupancy clearly labeled as simulated rather than live data, consistent with the system's explainability and honesty requirements. Overall, the results confirm that a single multi-agent pipeline, built from classical AI techniques, generalizes correctly across different real-world venues without venue-specific hardcoding.

CHAPTER 6

CONCLUSION AND FUTURE WORK

This project presents FlowGuard, a location-based multi-agent digital twin for crowd management that generalizes to any real-world venue rather than a single fixed site. By combining real OpenStreetMap geographic data, classical image-analysis-based capacity estimation, BDI-structured agents, a utility-based Coordinator, Dijkstra/A* routing, and CSP-based exit assignment, the system produces fully explainable crowd-management decisions without relying on machine learning or deep learning. The addition of graph-based risk-point detection and emergency-responder routing with ranked hospital selection extends the system beyond routine crowd routing into safety-critical decision support.

Testing across three real Tamil Nadu venues — Chepauk, Rajarathinam Stadium, and Jawaharlal Nehru International Stadium — demonstrated that the same agent pipeline correctly adapts to different real graphs, capacities, and risk profiles without venue-specific tuning, and that agent-coordinated rerouting measurably reduces evacuation time and peak crowd density compared to an unmanaged baseline.

As future work, the system can be extended with a transparent, non-ML demand-forecasting module driven by holiday, festival, and weather data; real visitor-facing SMS/app notifications; live road-traffic integration for responder routing; and broader validation against publicly reported real-world crowd-congestion incidents. Further work could also explore negotiation-based coordination between agents with competing goals (for example, Route Agent and a dedicated Parking Agent) to study more decentralized multi-agent consensus mechanisms.

CHAPTER 7

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